

Overview

Objective & key questions

Business Objective
Analyse sales trends, profitability, promotional performance, and customer demand to provide data-driven business recommendations.

Key business questions

- How has the product performed over time?

When was the product likely on promotion?

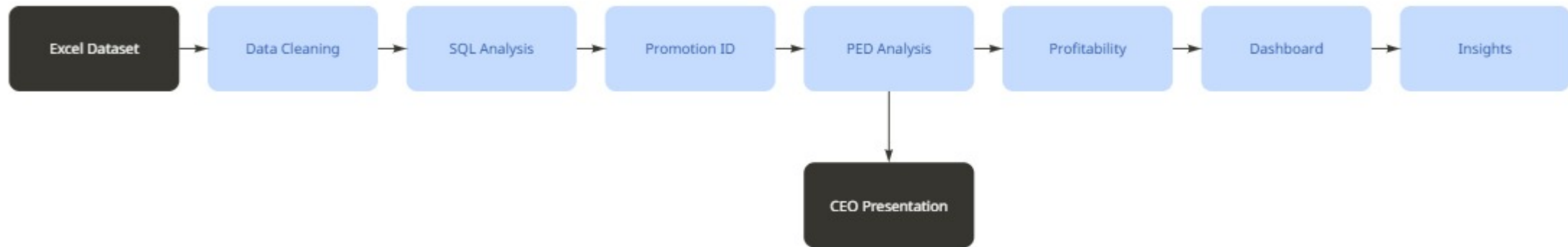
How do promotions affect sales & profit?

What is the Price Elasticity of Demand?

Does it perform better at promo prices?

Workflow

End-to-end analytics pipeline



Data Prep & SQL

Calculated metrics and SQL concepts

Calculated metrics

Total
Sales

Cost of
Sales

Quantity
Sold

Daily
Unit
Price

Gross
Profit

Gross
Profit %

GP per
Unit

Prev
Unit
Price

Prev Qty
Sold

Promotion
Status

SQL concepts

Common Table Expressions (CTEs)

YEAR()

CASE WHEN statements

MONTHNAME()

LAG() window functions

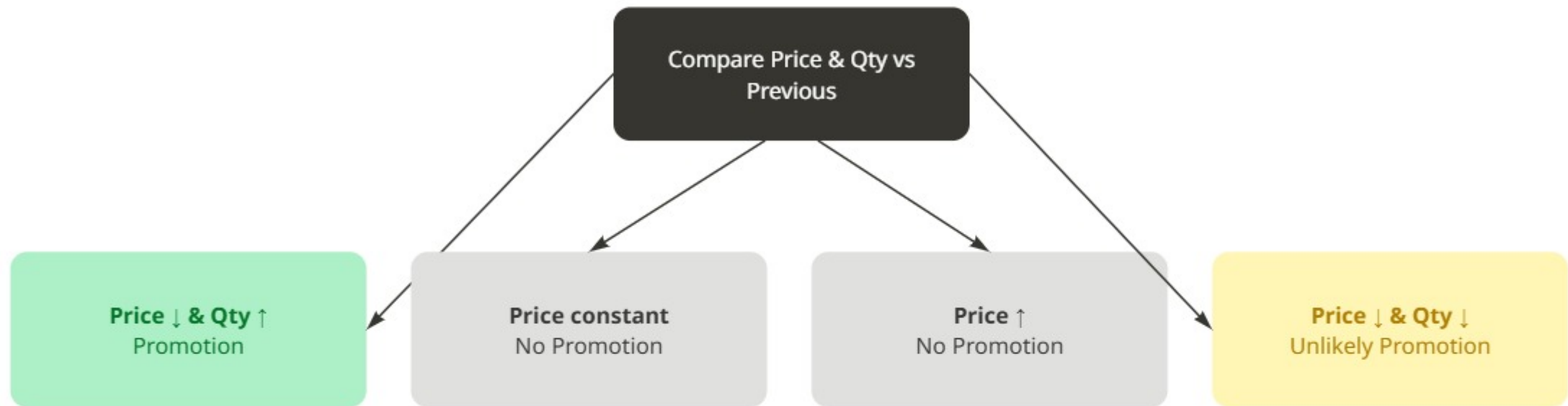
DAYNAME()

ROUND()

Aggregates: SUM, AVG, COUNT

Promotion Logic

Decision rules



Price Elasticity

Formulas & interpretation

$$\text{Sales Price / Unit} = \text{Sales} \div \text{Quantity Sold}$$

$$\text{Gross Profit / Unit} = (\text{Sales} - \text{Cost of Sales}) \div \text{Qty}$$

$$\% \Delta \text{ Quantity} = ((\text{New Qty} - \text{Old Qty}) \div \text{Old Qty}) \times 100$$

$$\% \Delta \text{ Price} = ((\text{New Price} - \text{Old Price}) \div \text{Old Price}) \times 100$$

$$\text{PED} = \% \Delta \text{ Quantity} \div \% \Delta \text{ Price}$$

Interpretation

PED > 1
Elastic Demand

PED = 1
Unit Elastic

PED < 1
Inelastic Demand

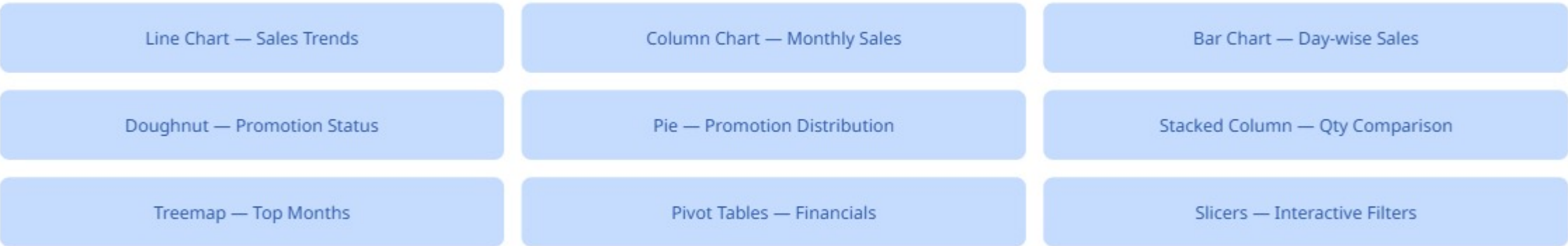
Dashboard Plan

KPIs, visuals & filters

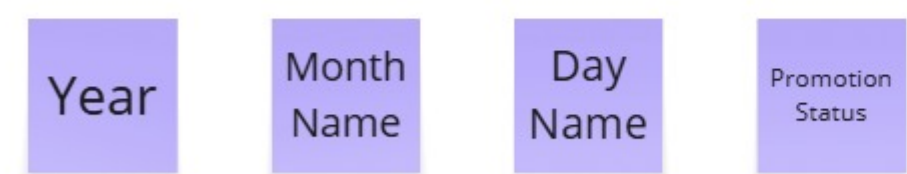
KPI cards



Recommended visualisations



Dashboard filters



Insights

Key findings

Promotions
increase
customer
demand

Customers
respond to
price
changes

Promo
pricing lifts
sales
volume

Some
periods
drive higher
profits

Pricing
strategy
shapes
profitability

Recommendations

Actions by function

Marketing

Implement targeted promotional campaigns

Promote products during low-performing periods

Sales Strategy

Optimise promotional pricing strategies

Focus on periods with high demand elasticity

Operations

Improve inventory planning during promotions

Monitor profitability & sales regularly

Reporting

Automate dashboard reporting

Use interactive dashboards for executives

Project Overview

Objective and key questions

Business Objective

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Key Business Questions

How has the product performed over time?

When was the product likely on promotion?

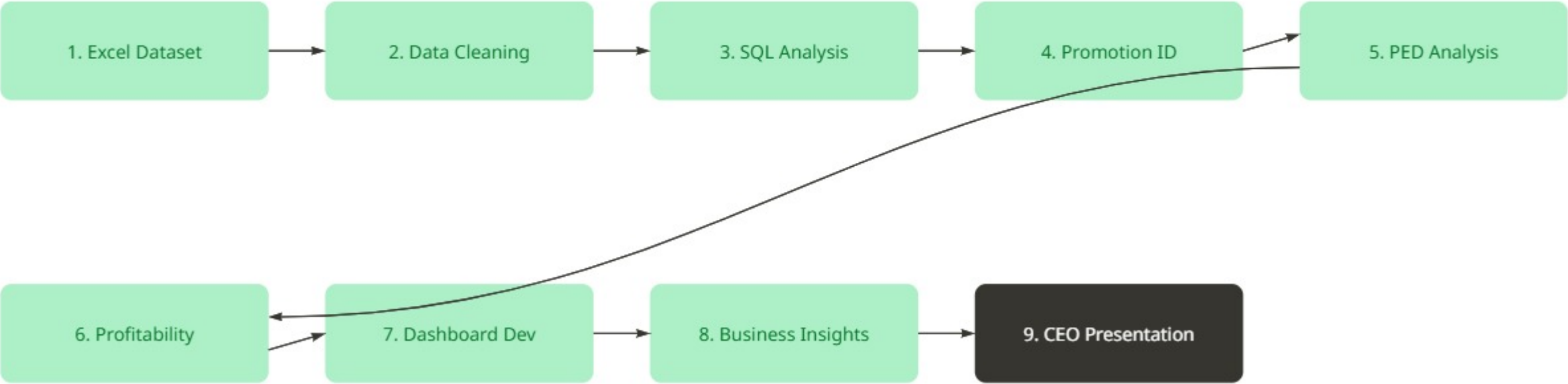
How do promotions affect sales and profitability?

What is the Price Elasticity of Demand during promotions?

Does the product perform better at promotional prices?

Project Workflow

End-to-end analytics pipeline



Stakeholders

Team

Q

Search by name or email

Online

P

phathutsdzo

Placeholder

Data Prep + SQL

Metrics calculated and SQL toolkit

Calculated Metrics



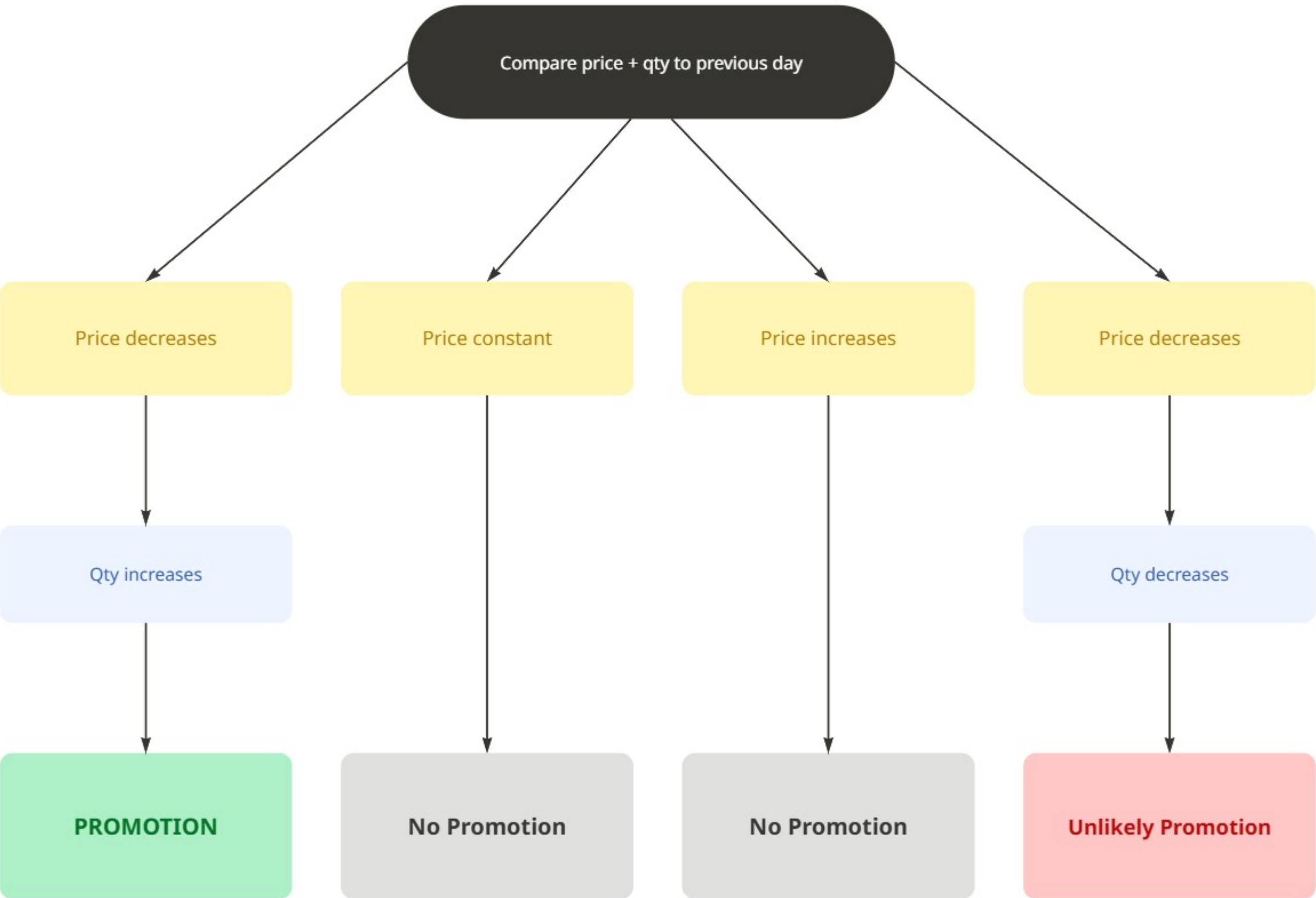
SQL Concepts

SQL Toolkit

```
1  -- Key SQL constructs used
2  WITH sales_cte AS (
3    SELECT date, product, sales, qty,
4           LAG(unit_price) OVER (ORDER BY date) AS prev_price,
5           LAG(qty) OVER (ORDER BY date) AS prev_qty
6    FROM sales
7  )
8  SELECT YEAR(date), MONTHNAME(date), DAYNAME(date),
9         ROUND(SUM(sales),2) AS total_sales,
10        CASE
11          WHEN unit_price < prev_price AND qty > prev_qty THEN 'Promotion'
12          WHEN unit_price = prev_price THEN 'No Promotion'
13          WHEN unit_price > prev_price THEN 'No Promotion'
14          ELSE 'Unlikely Promotion'
15        END AS promo_status
16  FROM sales_cte GROUP BY 1,2,3;
```

Promotion Logic

Decision tree for promotion status



Price Elasticity

Formulas and interpretation

Formulas

$\text{Sales Price per Unit} = \text{Sales} / \text{Quantity Sold}$

$\text{Gross Profit per Unit} = (\text{Sales} - \text{Cost of Sales}) / \text{Quantity Sold}$

$\% \text{ Change Quantity} = ((\text{New Qty} - \text{Old Qty}) / \text{Old Qty}) \times 100$

$\% \text{ Change Price} = ((\text{New Price} - \text{Old Price}) / \text{Old Price}) \times 100$

$\text{PED} = \% \text{ Change in Quantity} / \% \text{ Change in Price}$

Interpretation

**PED > 1 Elastic
Demand**

**PED = 1
Unit Elastic**

**PED < 1 Inelastic
Demand**

Dashboard Planning

KPIs, visualisations, and filters

KPI Cards

Total Sales

Total Gross Profit

Total Quantity Sold

Avg Unit Price

Promotional Days

Recommended Visualisations

Line:
Sales Trends

Column:
Monthly Sales

Bar:
Day-wise Sales

Doughnut:
Promo Status

Pie: Promo
Distribution

Stacked
Col: Qty Compare

Treemap:
Top Months

Pivot:
Financials

Slicers:
Filters

Dashboard Filters

Year

Month Name

Day Name

Promotion Status

Business Insights

Key findings from analysis

Promotions
increase
customer
demand.

Customers are
responsive to
price changes.

Promotional
pricing
positively
impacts sales
volume.

Some periods
generate
significantly
higher sales
and profits.

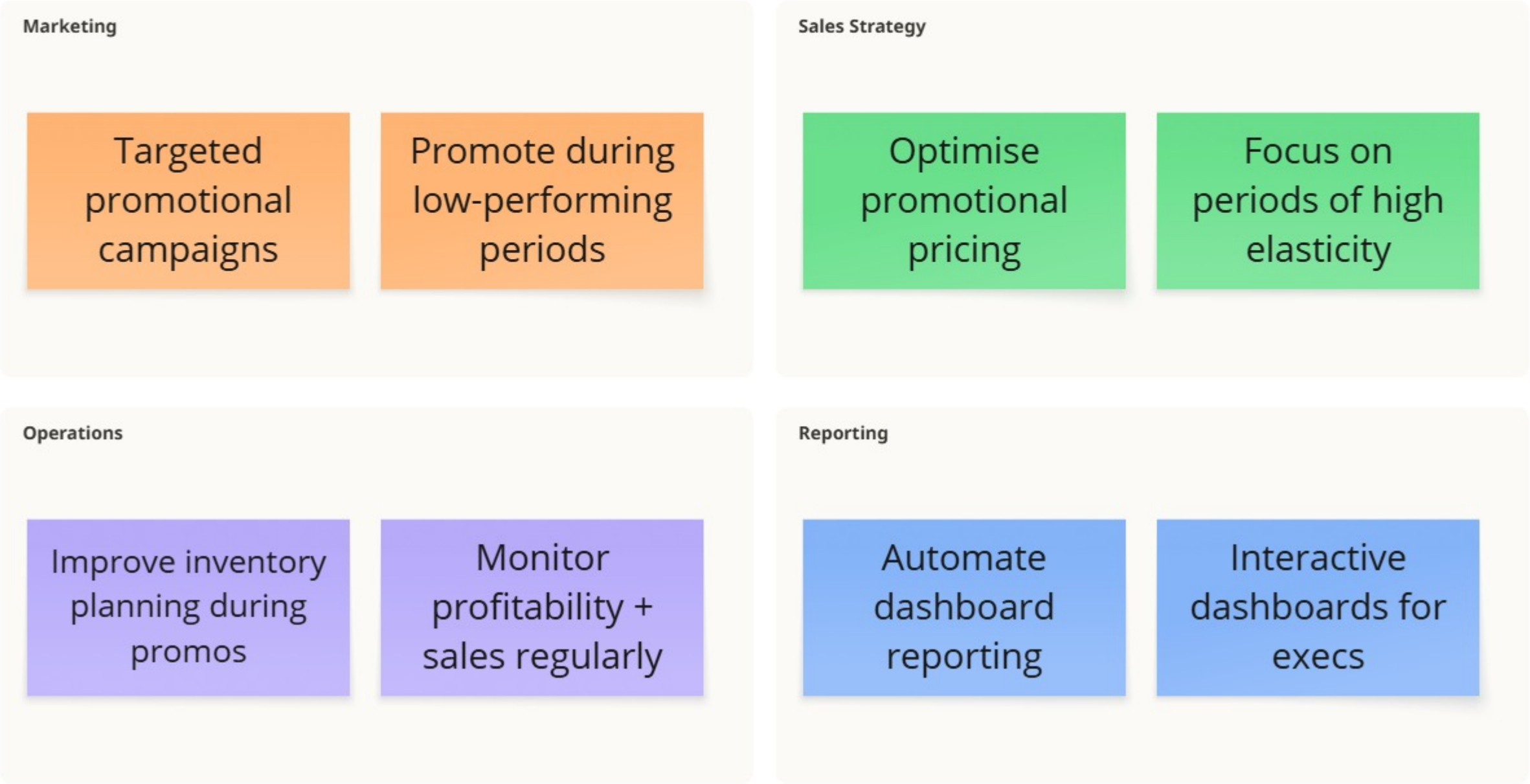
Pricing
strategies
influence
profitability.

Dot voting



Recommendations

Action areas for the CEO



Next step: Present findings and recommendations to the CEO.